Ka-Chow Net: The Radiator Springs Network

Introduction

Radiator Springs is buzzing: Lightning McQueen has rolled back into town for off-season training with Doc Hudson before the next Piston Cup qualifier, and the whole town is gearing up to host a live "Radiator Springs Grand Prix" viewing event. To keep the town coordinated (e.g. make sure every resident can stream McQueen's practice laps, order tires, email Luigi about whitewalls etc), the town council has commissioned a unified digital network, Ka-Chow Net, connecting every business and service in town.

You have been assigned to design and implement the base networks for the seven core units of Radiator Springs. The estimated number of hosts/agents at each unit is given below:

Unit	Abbreviation	Role in Town	Hosts
Sheriff Station	SS	Town control	260
Luigi Casa-Della Tires	LCT	Tires shop	220
Doc-Hudson Clinic	DC	Medical center	180
Flo V8 Café	FVC	Fuelling station	150
Ramone Body Art	RBA	Paint shop	90
Mater Tow Yard	MTY	Towing and repair	90
Wheel Well Motel	WWM	Roadside motel	40

Requirements

Topology Design & Server Setup

Core Nodes: Every unit is represented by a router. Each unit has its own LAN switch, exactly two PCs representing all hosts/agents, and one dedicated network printer.

Isolated Link: Wheel Well Motel (WWM) is only connected to Flo V8 Café (FVC)

Downtown Hub: Sheriff Station (SS), Luigi Casa-Della Tires (LCT), Doc-Hudson Clinic (DC), Flo V8 Café (FVC), and Mater Tow Yard (MTY) are all connected to one another via a shared central switch. SS and FVC also have an additional direct point-to-point link because of an old dispatch radio line between the Sheriff and Flo's that never got decommissioned.

The Gossip Loop: A network loop connects the three units that keep the town informed: Sheriff Station (SS) is directly connected to Ramone Body Art (RBA), RBA is connected to Doc-Hudson Clinic (DC), and DC is connected back to SS.

Local Email Servers: Every unit runs its own local email server for internal communication, using hostnames of the form *mail.sheriff.rs, mail.luigi.rs, mail.doc.rs,* etc.

Cross-Domain Email: Sheriff Station (*mail.sheriff.rs*) and Flo V8 Café (*mail.flo.rs*) must be configured so that at least two user accounts on each server can send and receive email with the other domain in both directions (e.g. *sheriff1@sheriff.rs can send to and receive mail from flo1@flo.rs*).

Dedicated Web Servers: Only two units have dedicated web servers:

1. Sheriff Station: *www.sheriff.radiatorsprings.nyc*
2. Flo V8 Café: *www.flo.radiatorsprings.nyc*

Configure them so every device across all seven units can open both sites by domain name.

Central Directory: Only Sheriff Station (SS) hosts the central DNS server. Configure it so every device in every unit uses it for domain name resolution.

Addressing

The base network address is derived from the 1st group member's student ID.

Take the last four digits of the student ID and split them into two groups of two digits.

- First pair → first octet *[Note: If the first pair is 00, use 10 instead]*
- Second pair → second octet

The third and fourth octets are 0, and the prefix mask is /16.

Example: Suppose the student ID is 20201002. The last four digits are 1002. Splitting gives 10 and 02, so the base network address is 10.2.0.0/16.

Subnet this base network using VLSM, allocating address blocks in the following order: **SS, DC, FVC, LCT, RBA, MTY, WWM**, and finally all point-to-point/inter-router links. Show the complete VLSM tree in your submission.

All PCs receive their IP addresses dynamically. Servers, routers, and printers are all configured with static/manual addressing.

DHCP Configuration

- SS's router acts as the DHCP server for the three Gossip Loop units: SS, RBA, DC.
- LCT and FVC each run their own dedicated local DHCP server for their own unit.
- WWM has no DHCP server of its own and uses FVC's DHCP server, since its only path to anywhere is through FVC.
- MTY has no local DHCP server. Instead, MTY's router must be configured to relay DHCP requests to LCT's DHCP server using an IP helper address, since LCT's server is not on MTY's local segment.

Routing

- **Dynamic Routing:** SS, RBA, and DC share their routing tables with one another using RIPv2 (the Gossip Loop).
- **Full Awareness via Redistribution:** SS's router must be aware of every network in the topology, some learned statically and some dynamically. SS must redistribute its static routes into RIPv2 so that RBA and DC automatically learn about the LCT, FVC, MTY, and WWM networks without being configured with any static routes of their own.
- **Default Gateway:** WWM can reach the rest of the network only through FVC; configure a default (static) route on WWM's router.
- **Static Mapping:** LCT and MTY must each be configured with static routes to reach every external network — no default routes are permitted on these two routers.
- **Exit-Interface Routing:** FVC must reach every other network using exit-interface static routes (the route statements must name only the outgoing interface, not a next-hop IP).
- **Redundancy (Recursive Routing):** If the direct SS–FVC link fails, FVC must reach the SS network through an alternate path via LCT, configured as a recursive static route (i.e. pointing to an intermediate router's address, not directly to an interface).
- **Redundancy (Floating Static Route):** LCT must also keep a floating static backup route to the SS network via DC. This floating route must sit in LCT's routing table but only become active if the primary route toward SS fails.

Testing & Verification

- After configuration, every device across all seven units must be able to ping each other.
- You must demonstrate, by shutting down the relevant primary interface(s), that FVC's recursive backup route and LCT's floating static route each take over correctly, and that traffic is restored end-to-end.
- You must demonstrate two-way email exchange between the *sheriff.rs* and *flo.rs* domains, and confirm both *www.sheriff.radiatorsprings.nyc* and *www.flosv8.radiatorsprings.nyc* open from a PC in every unit (this is for simulating the equivalent of the whole town watching the Piston Cup qualifier live).

You are allowed to make any valid and necessary networking assumptions not explicitly stated above, provided they do not conflict with, remove, or simplify any requirement listed here.

Deliverables

1. **Work Distribution** — clearly state which group member configured which unit(s) and services.
2. The completed **.pkt/.pka** Packet Tracer file.
3. A picture of the full network topology diagram, clearly labelled, with the network address of every link/segment shown as a note.
4. A PDF report containing:
   - The complete VLSM tree.
   - The full IP addressing table (unit, device, IP address, subnet mask, default gateway).
   - All router configuration commands (running-config or equivalent) for all seven routers.
   - DHCP, DNS, web, and email server configuration details/screenshots.
   - Screenshots demonstrating successful pings, the two failover tests, and the cross-domain email exchange.